

Q1. Tell me about yourself.

A1. Answer:

Hi, my name is Chandrashekhar, and I'm currently working as an Azure Data Engineer on a real production project. My primary responsibility is to design, develop, automate, and maintain data pipelines that move data from different source systems into Azure for analytics and reporting. I have hands-on experience with Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, PySpark, Delta Lake, Azure Synapse Analytics, and Azure SQL Database. I also work extensively with SQL, Python, ETL/ELT pipelines, incremental data loading, data transformation, monitoring, and troubleshooting. In my current project, we ingest data from multiple sources such as SQL databases, APIs, and files. ADF is mainly used for orchestration and ingestion, while Databricks and PySpark are used for large-scale transformation. We store data in different layers in ADLS Gen2 and use Delta Lake for reliable processing and analytics. I work closely with business analysts, application teams, developers, QA, and reporting teams to understand requirements and build scalable data solutions. I also spend time troubleshooting pipeline failures, optimizing Spark jobs and SQL queries, handling data-quality issues, and improving pipeline performance and reliability. Overall, I enjoy solving real-world data engineering problems and building scalable, reliable, and maintainable data platforms on Azure.

📌 Interview Follow-up Questions

- Which Azure services have you worked with most?
- Can you explain your current project architecture?
- What is your role in the project?
- What was the biggest challenge you faced?

Q2. What are your roles and responsibilities in your current project?

A2. Answer:

In my current project, I work as an Azure Data Engineer, and my main responsibility is to build and maintain reliable data pipelines for our analytics platform. I use Azure Data Factory to develop and orchestrate ETL/ELT pipelines. This includes activities such as Copy Data, Lookup, Stored Procedure, Execute Pipeline, and Databricks activities. For storage, we use Azure Data Lake Storage Gen2, where we organize data into Raw, Processed, and Curated layers. For large-scale transformations, I work with Azure Databricks and PySpark. I handle data cleansing, joins, aggregations, deduplication, data validation, and business-rule implementation. I also work with Delta Lake for reliable data processing and maintaining curated datasets. On the analytics side, I work with Azure Synapse Analytics and SQL to prepare data for reporting and analytical workloads. Another important part of my role is implementing incremental data loads. Depending on the source, we use watermark columns, timestamps, or other mechanisms to identify new and updated records instead of processing the entire dataset every time. I'm also responsible for pipeline monitoring and troubleshooting. When a pipeline fails, I analyze the error, identify the root cause, fix the issue, and make sure the solution does not introduce similar failures in future runs. Apart from development, I participate in requirement discussions, code reviews, testing, documentation, and performance/cost improvement activities.

📌 My Core Responsibilities

- ADF pipeline development and orchestration
- ADLS Gen2 data lake design and organization
- Databricks + PySpark transformations
- Delta Lake processing
- SQL and Synapse analytics
- Incremental and full data loads
- Monitoring, troubleshooting, and RCA
- Data quality, testing, deployment, and documentation

Q3. Walk me through your day-to-day activities.

A3. Answer:

My day usually starts by checking the status of our production data pipelines and scheduled jobs. I review Azure Data Factory and Databricks monitoring to make sure the previous day's or overnight pipelines completed successfully. If there are any failures, I prioritize them based on business impact. After that, I usually work on development or enhancement activities. For example, I may create a new ADF pipeline, implement an incremental load, modify a PySpark transformation, or add a new dataset to the data lake. I also spend time working with SQL and PySpark for data validation. If there is a mismatch between source and target data, I compare record counts, business keys, timestamps, and transformation logic to identify where the issue occurred. During the day, I participate in stand-up meetings and requirement discussions with business analysts, developers, QA, and other data engineering team members. If we have a production issue, I work on troubleshooting and root cause analysis. It could be a source connectivity issue, schema change, bad data, Spark job failure, pipeline timeout, or performance problem. I also work on optimizing pipelines by reducing unnecessary data movement, improving SQL queries, optimizing Spark transformations, managing partitioning, and avoiding unnecessary full loads. Before completing a development task, I perform testing, validate the output, and make sure the changes are ready for deployment.

☒ Typical Day Flow



☒ Important:

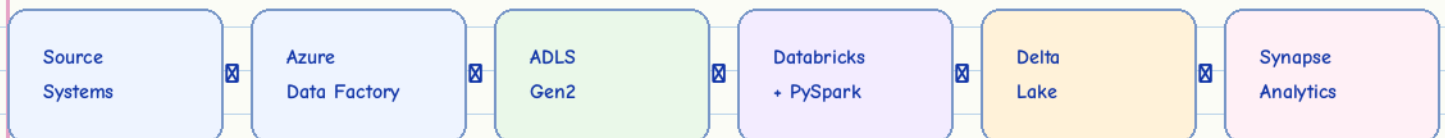
A strong answer connects your daily activities to production reliability: monitor ☒ develop ☒ validate ☒ troubleshoot ☒ deploy.

Q4. Walk me through your project.

A4. Answer:

Sure. I'm currently working on a cloud-based data engineering and analytics platform on Azure. The main objective of the project is to collect data from multiple source systems, process and transform that data, and provide trusted datasets for reporting and business analytics. Our data comes from sources such as Azure SQL or SQL Server databases, REST APIs, and flat files such as CSV and JSON. At the ingestion layer, we use Azure Data Factory. ADF connects to source systems, extracts the data, and loads it into Azure Data Lake Storage Gen2. We maintain different layers in the data lake. The Raw layer contains source data, the Processed layer contains cleaned and standardized data, and the Curated layer contains business-ready datasets. For large-scale transformations, we use Azure Databricks with PySpark. We perform cleansing, joins, aggregations, deduplication, validation, and business transformations. We use Delta Lake for important datasets where we need reliable table management and transactional consistency. For analytical workloads, curated data is made available through Azure Synapse Analytics for reporting and analytics. A key part of the project is incremental processing. Instead of loading the complete source every day, we identify new or modified records using a watermark such as a last-modified timestamp or incremental key. We also have monitoring and error-handling mechanisms. We track pipeline status, execution time, errors, and record counts, and investigate failures through root cause analysis. From a security perspective, we follow least-privilege access, managed identities or service principals where appropriate, secure connectivity, and controlled access to data lake resources.

📁 Project Architecture



📁 Core Data Flow

Source Systems → ADF → ADLS Gen2 → Databricks / PySpark → Delta Lake → Synapse Analytics → Reporting

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Q5. What was your contribution to the project?

A5. Answer:

My major contribution was around building and maintaining the data pipelines end-to-end. I worked on ADF pipelines for source ingestion, implemented incremental loading logic, developed PySpark transformations in Databricks, and prepared curated datasets for downstream consumption. I was also involved in troubleshooting production failures and data-quality issues. For performance improvements, I worked on optimizing SQL queries, Spark transformations, partitioning strategies, and reducing unnecessary data processing. I participated in requirement discussions and worked with QA and reporting teams during testing and release activities. So, I would say my contribution covered the complete data engineering lifecycle from ingestion to transformation, validation, optimization, and production support.

☒ Challenges & How I Approach Them

Pipeline failure

Check activity error ☒ identify root cause ☒ fix ☒ rerun safely

Data mismatch

Compare counts, keys, timestamps, and transformation logic

Slow Spark job

Check partitions, joins, shuffles, file sizes, and execution plan

Schema change

Validate source change ☒ update mapping/transformation ☒ test downstream

☒ Follow-up Questions

- Why did you choose ADF for orchestration?
- Why did you use Databricks instead of doing everything in ADF?
- How did you implement incremental loading?
- How did you handle schema changes?
- How did you optimize Databricks jobs?
- How did you monitor production pipelines?
- How did you secure your Azure data platform?
- Tell me about the biggest production issue you solved.